

DevOps Engineering Foundation - 6-Week Intensive Bootcamp

WEEK 5: Infrastructure as Code & Orchestration (Start Final Project)

Learning Outcomes:

- Understand Infrastructure as Code (IaC) principles
- Provision infrastructure using Terraform
- Manage Kubernetes clusters and deploy applications
- Begin architecting the final capstone project

Resources:

Infrastructure as Code:

- [What is Infrastructure as Code?](#) by IBM Technology (8 min)
- [IaC Best Practices](#)

Terraform:

- [Terraform Course - Automate your AWS Infrastructure](#) by freeCodeCamp (2 hours)
- [Terraform in 100 Seconds](#) by Fireship (2 min)
- [Complete Terraform Course](#) by DevOps Directive (2 hours)
- [Terraform Documentation - Get Started](#)
- [Terraform Best Practices](#)

Kubernetes:

- [Kubernetes Tutorial for Beginners - Full Course](#) by TechWorld with Nana (4 hours)
- [Kubernetes Crash Course for Absolute Beginners](#) by TechWorld with Nana (1 hour)
- [Kubernetes Documentation - Tutorials](#)
- [Play with Kubernetes](#) - Interactive playground

Container Orchestration:

- [Kubernetes vs Docker Swarm](#) by IBM Technology (7 min)
- [Kubernetes Basics](#)

Week 5 Task:

Infrastructure Provisioning with Terraform

Create reusable Terraform modules:

1. Provision AWS infrastructure:
 - VPC with public and private subnets
 - EC2 instances with Auto Scaling Group
 - Security groups with least privilege access
 - S3 bucket for static assets
 - RDS database instance
2. Use variables and outputs effectively
3. Implement remote state management
4. Create separate environments (dev, staging, prod)
5. Deploy a simple Kubernetes cluster (EKS or Minikube locally)
6. Deploy a containerized application to Kubernetes

Deliverables:

- Terraform code with modules structure
- State files and backend configuration
- README with usage instructions
- Architecture diagram
- Cost estimation documentation

FINAL PROJECT BEGINS: Plan and design your capstone project architecture (to be built in Week 6)